

Retail Sales Performance Analysis

Task 1 : Pre-processing (Data cleaning):

Checked for all the null values, data type. Changed the date data type to date format, later added a month column to the dataset.

```
: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype  
---  --
 0   Order_ID        1500 non-null   object  
 1   Date             1500 non-null   object  
 2   Customer_ID      1500 non-null   object  
 3   Gender           1500 non-null   object  
 4   Age              1500 non-null   int64   
 5   City             1500 non-null   object  
 6   Category         1500 non-null   object  
 7   Quantity         1500 non-null   int64   
 8   Unit_Price       1500 non-null   int64   
 9   Sub_Category     1500 non-null   object  
10   Total_Amount     1500 non-null   int64   
dtypes: int64(4), object(7)
memory usage: 129.0+ KB

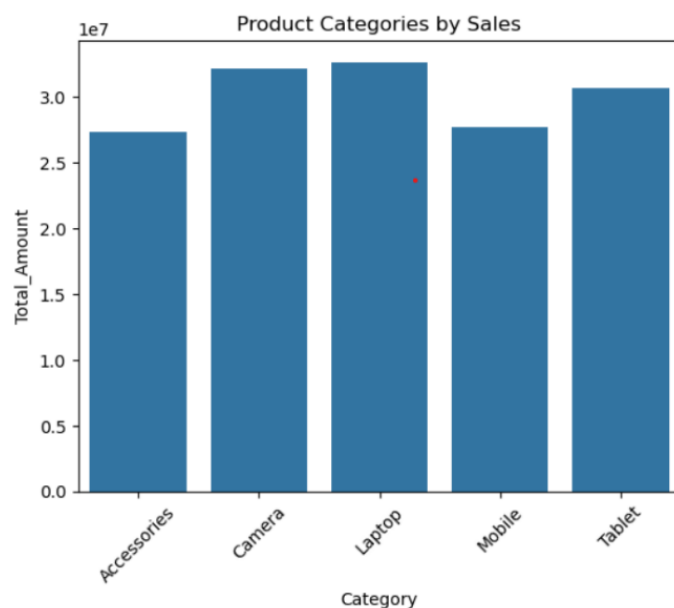
: df['Date'] = pd.to_datetime(df['Date'], format='%d-%m-%Y')    ## Convert Date Column to Datetime

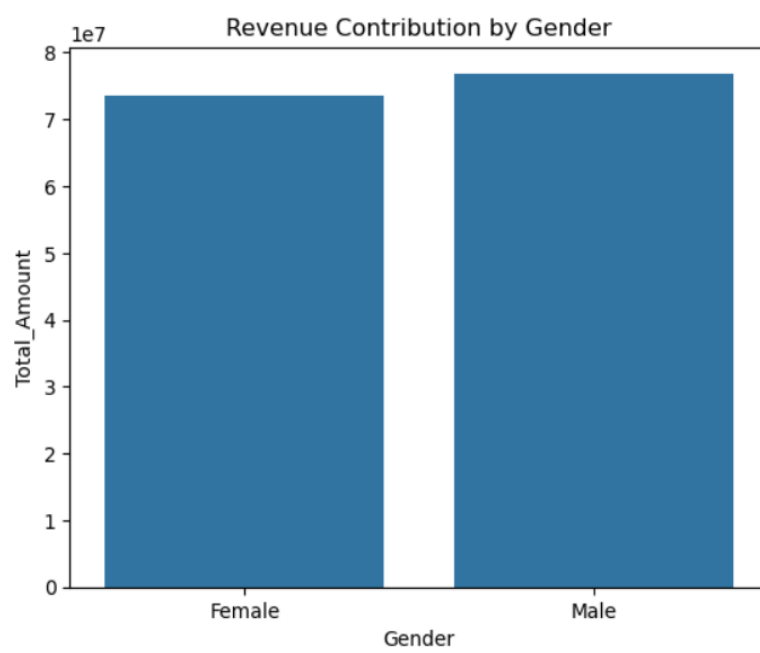
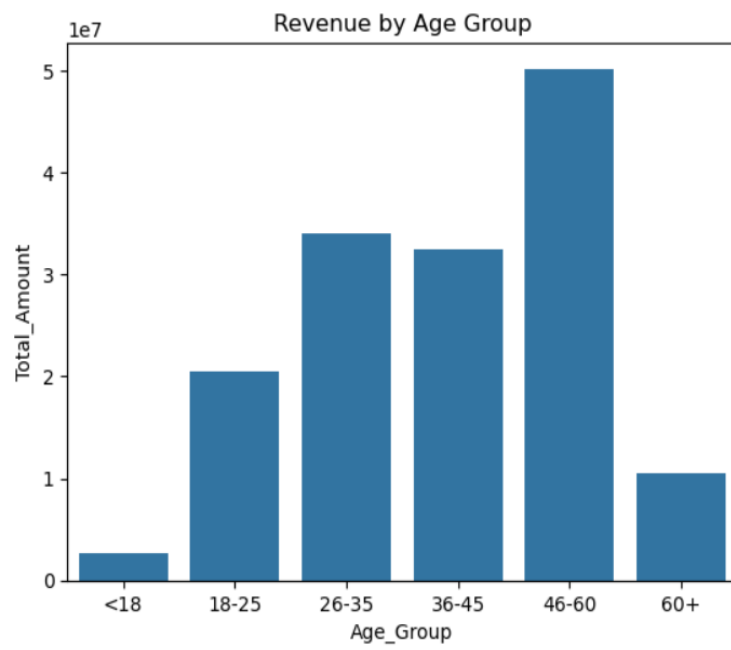
: df['Month_Name'] = df['Date'].dt.month_name()                  ## Create Month_name Column
```

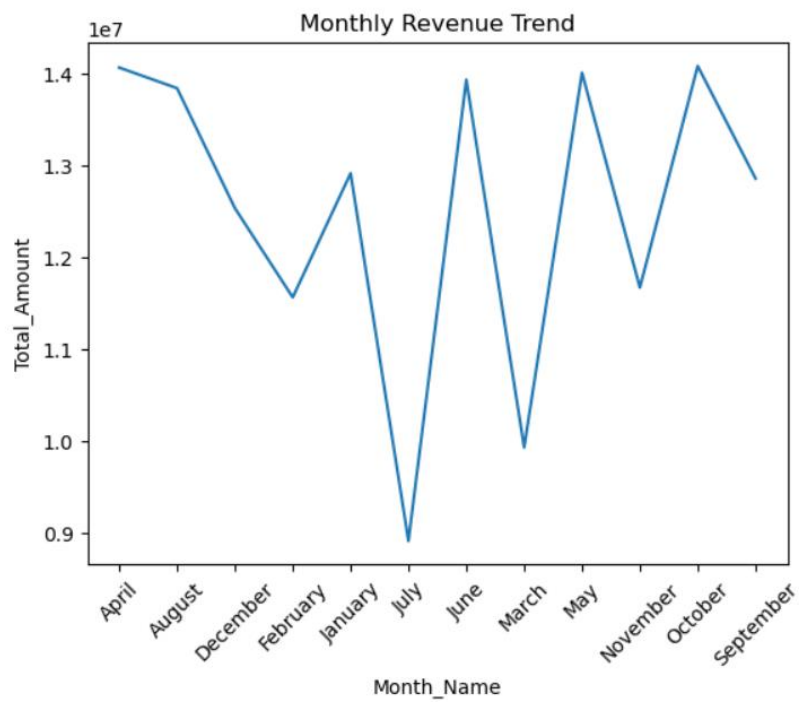
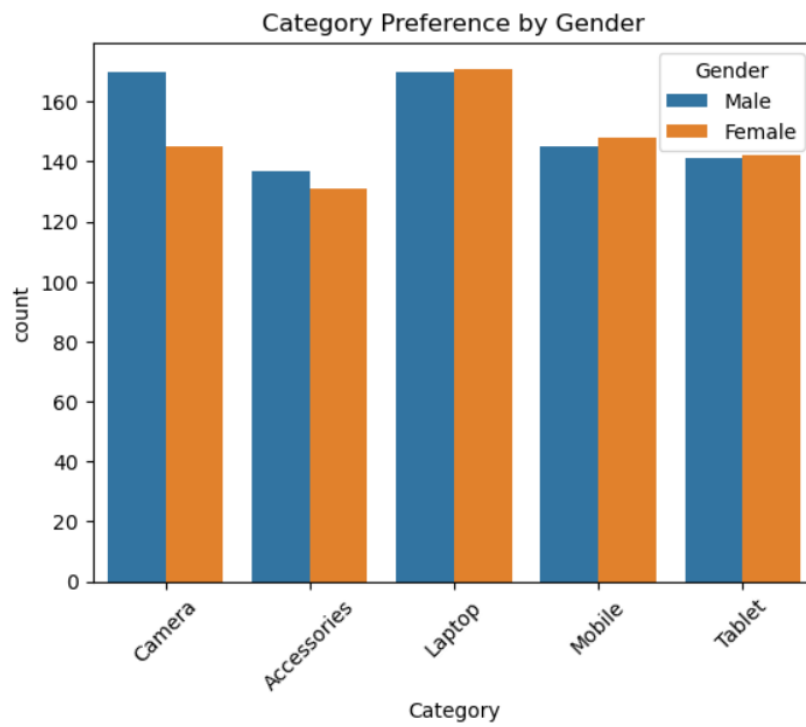
Task 2 : Visualisation

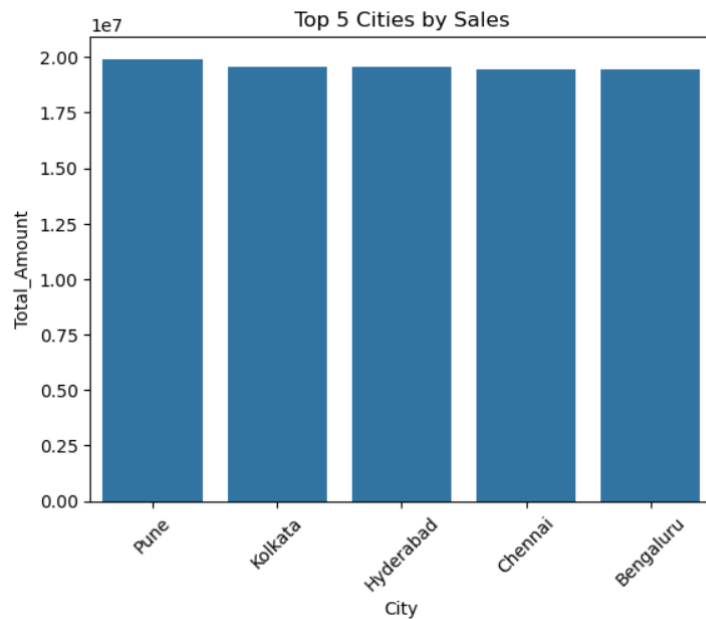
Performed data visualisation using python libraries. Imported the necessary library.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```









- Laptop category generates highest revenue.
- Age group 46 - 60 spends the most.
- Males purchase more than female.
- Accessories sell consistently across genders.
- Sales peak during April, May, June.
- Pune

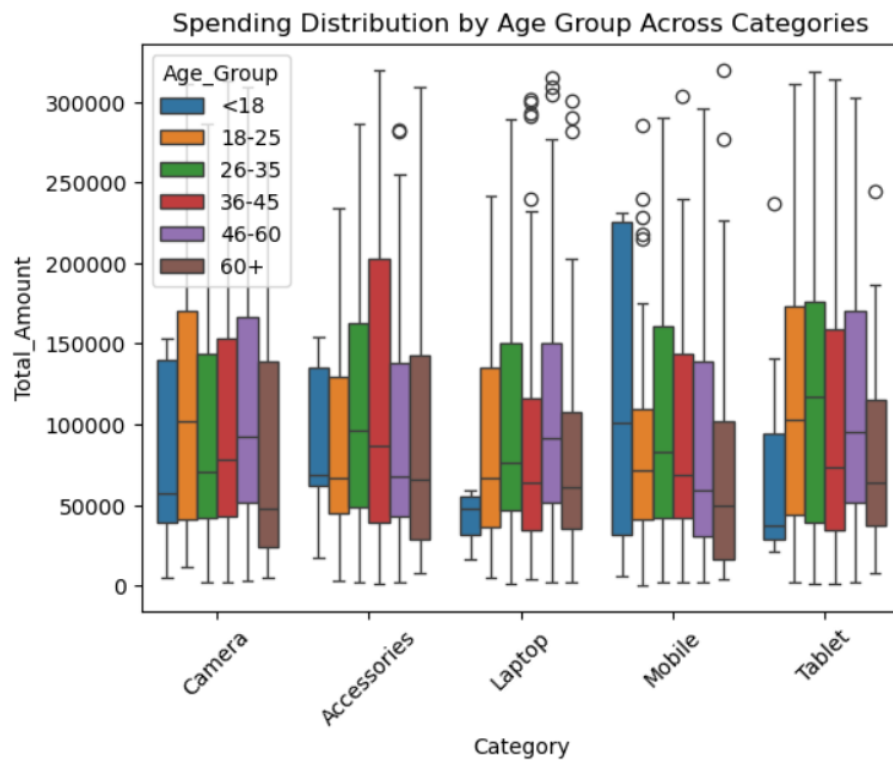
Task 3:

Study how customer age significantly impact average spending per transaction across different product categories.

Calculate Average order value :

```
## Calculate Average Spending
avg_spending = df.groupby(['Category', 'Age_Group'])['Total_Amount'].mean().reset_index()
```

Plot the findings in a box plot, which is a best visualisation for comparison. It shows spending variability.



ANOVA test was conducted to identify the statistical significance.

```
from scipy.stats import f_oneway

for category in df['Category'].unique():
    data = df[df['Category'] == category]

    groups = [group['Total_Amount'].values
               for name, group in data.groupby('Age_Group')]

    f_stat, p_value = f_oneway(*groups)

    print(f"{category} → p-value: {p_value}")
```

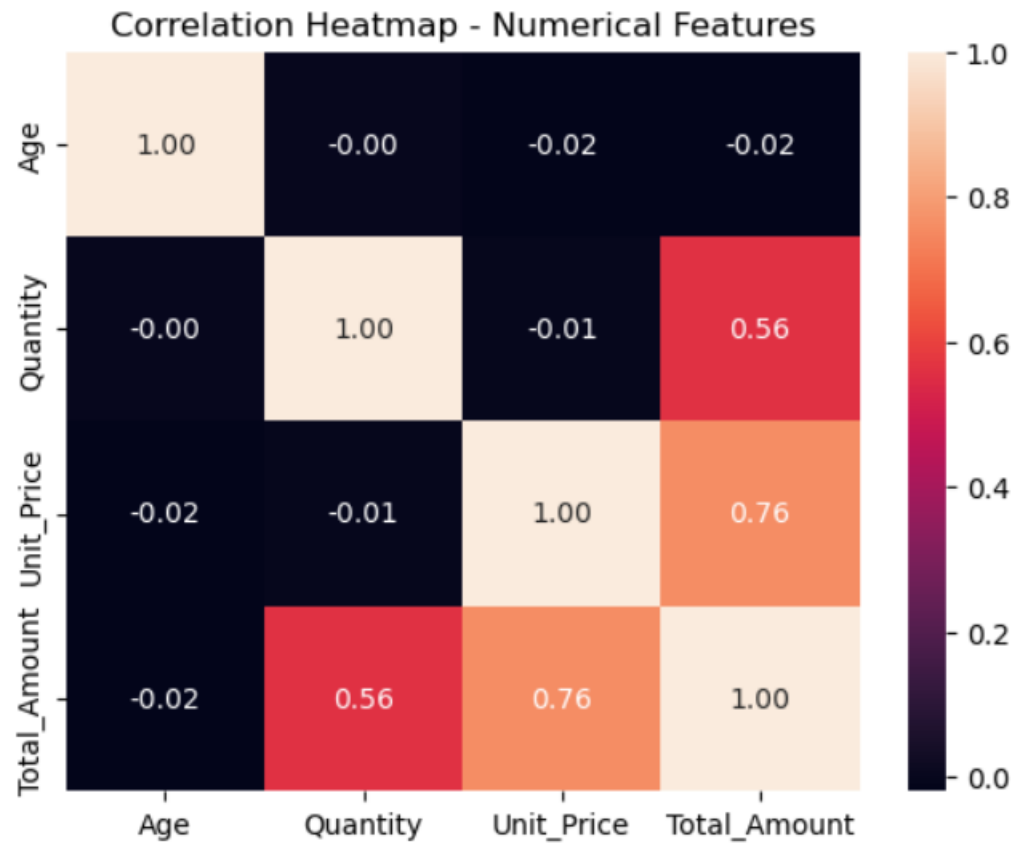
```
Camera → p-value: 0.2487965052454601
Accessories → p-value: 0.5220609408507102
Laptop → p-value: 0.17859278450843027
Mobile → p-value: 0.6011023072278802
Tablet → p-value: 0.21557149565053424
```

p-value > 0.05 which means No significant difference between age and spending. age alone is not a strong predictor of spending behavior.

Correlation study : Heatmap to visualize correlation between numerical features.

```
numeric_data = df.select_dtypes(include=['number'])
corr_matrix = numeric_data.corr()
```

```
plt.figure()
sns.heatmap(corr_matrix, annot=True, fmt=".2f")
plt.title("Correlation Heatmap - Numerical Features")
plt.show()
```

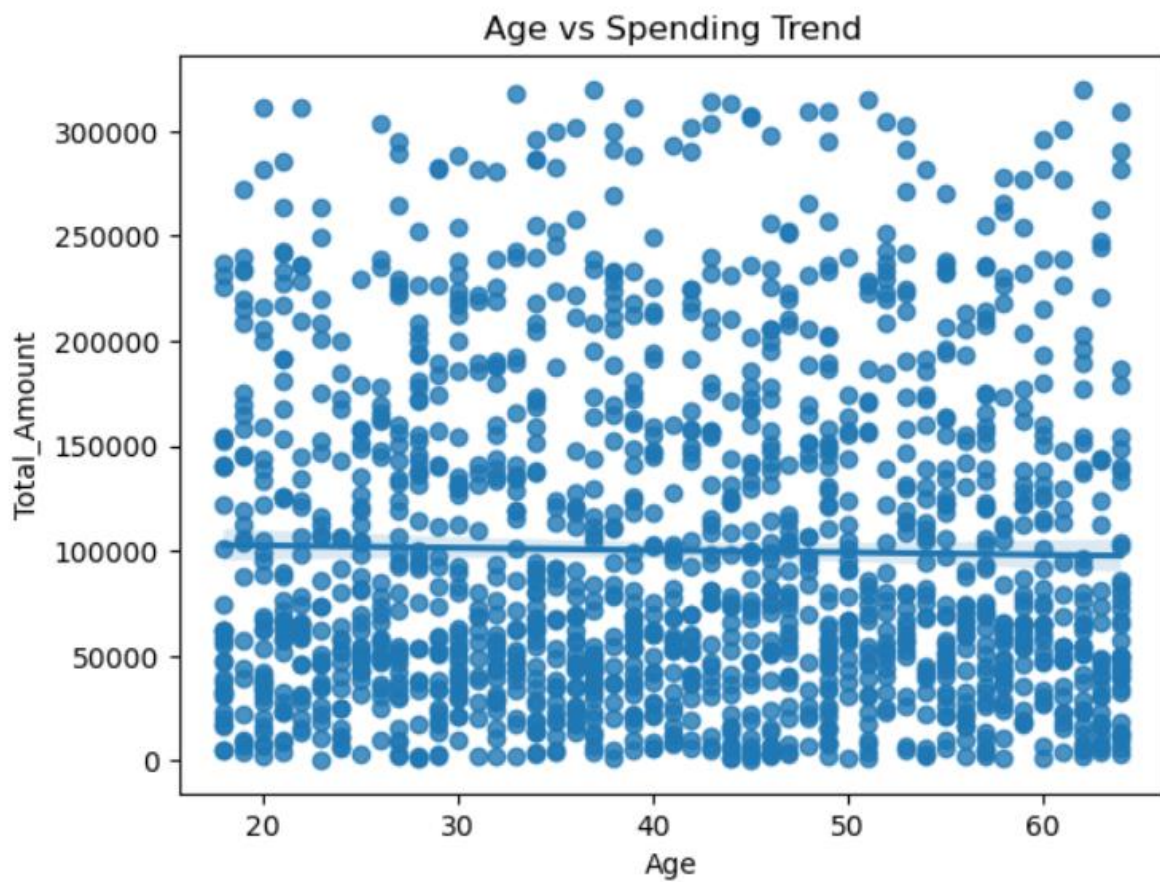


Calculate the correlation coefficient and visualize the relationship.

```
df[['Age', 'Total_Amount']].corr()
```

	Age	Total_Amount
Age	1.000000	-0.019229
Total_Amount	-0.019229	1.000000

Visualize Correlation (Scatter Plot)



Straight line indicate there is no correlation between age and spending.

The correlation heatmap shows that Unit Price has a strong positive relationship with Total Amount, indicating that higher-priced products contribute significantly to revenue, while Age shows only a weak relationship with spending.